Income Prediction: Classification and Feature Selection
Data Preprocessing & EDA

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Dataset Overview

In our project, we use the Census-Income (KDD) dataset from the UCI Machine Learning Repository. The dataset contains information related to demographics and employment, based on survey data collected in the US in 1994 and 1995. From the original 299,285 records, we selected 215,574 records of individuals aged 18 or older for our analysis. We also excluded the instance_weight feature, which represents the survey weight. This left 40 input features, consisting of 7 numerical features and 33 categorical features.

The original data is available at the Census-Income (KDD) dataset in the UCI Machine Learning Repository. As the data size is large, we did not include the original data in the submission zip and the Git repository. The first code cell in the notebook executes “src/data_loader.py”, and then it automatically starts downloading the data from UCI. In this process, it combines two files (census-income.data and census-income.test), and then reduces the data so that it only contains people aged 18 or older. After that, it produces “data/processed/census_income_adults.csv”. Because of this, the first execution requires an internet connection.

We will classify whether an individual earns at least 50,000 USD per year. Among the selected records, 18,566 (8.61%) are in the high-income class, and 197,008 (91.39%) are in the other class. Because we found that the class distribution is highly imbalanced, we will use Average Precision (AP) as the main evaluation metric in the classification experiments.

Exploratory Data Analysis

In EDA, we examined feature distributions, missing values, outliers, and relationships between features. For numerical features, we created histograms and box plots by income class. For the four features related to money, 85% to 97% of the values were 0, and some values were extremely high. To make the distributions easier to check, we used a log1p transformation only for visualization. Moreover, in the preprocessing, we did not use any data augmentation such as SMOTE. Considering class imbalance, we mainly used AP as an evaluation index, and we maintained the class distribution by a stratified split. Also, we compared two cases: with and without class weights for training.

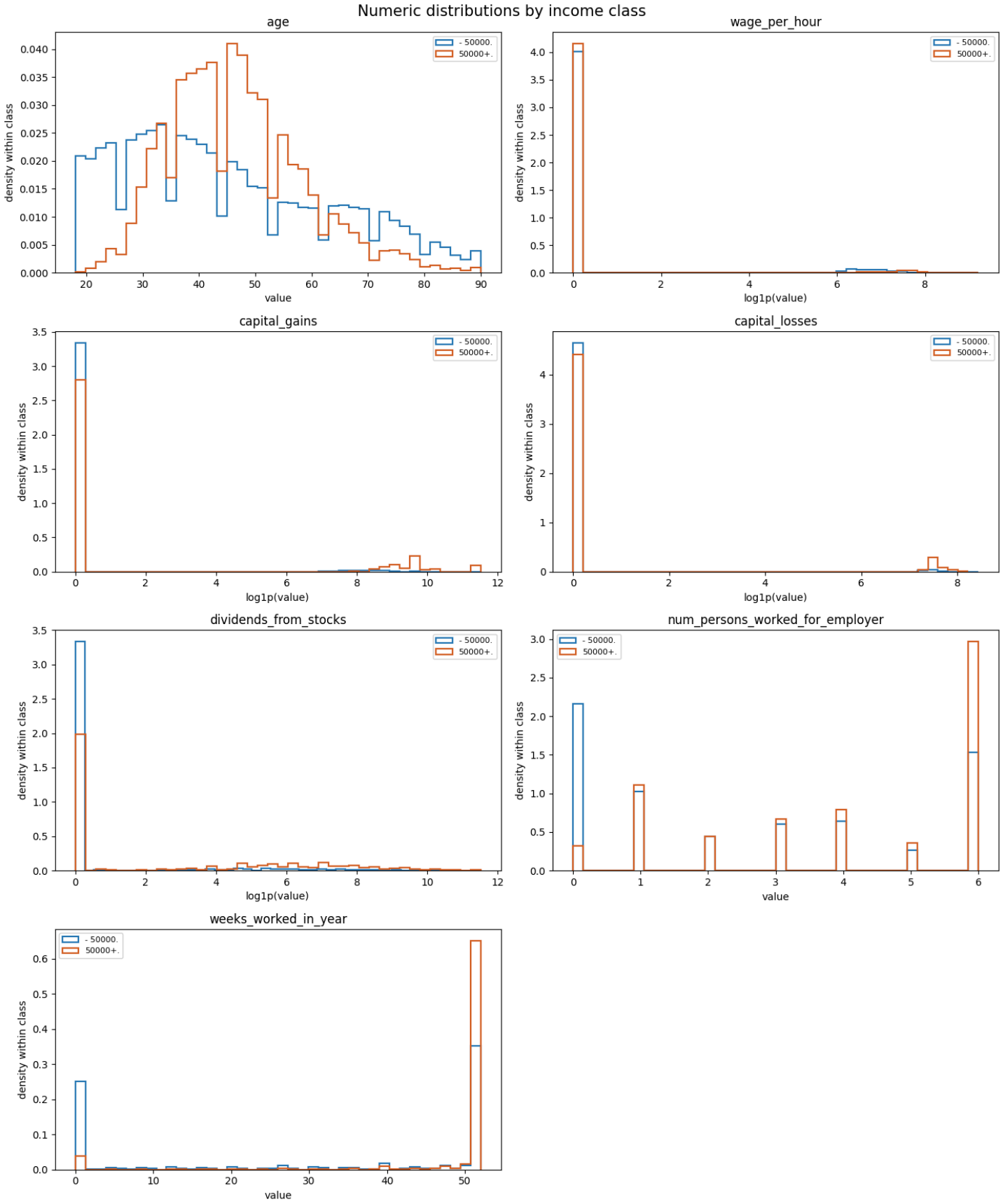


Figure 1. Histograms of numerical features by income class. The four money-related features are shown using a log1p scale.

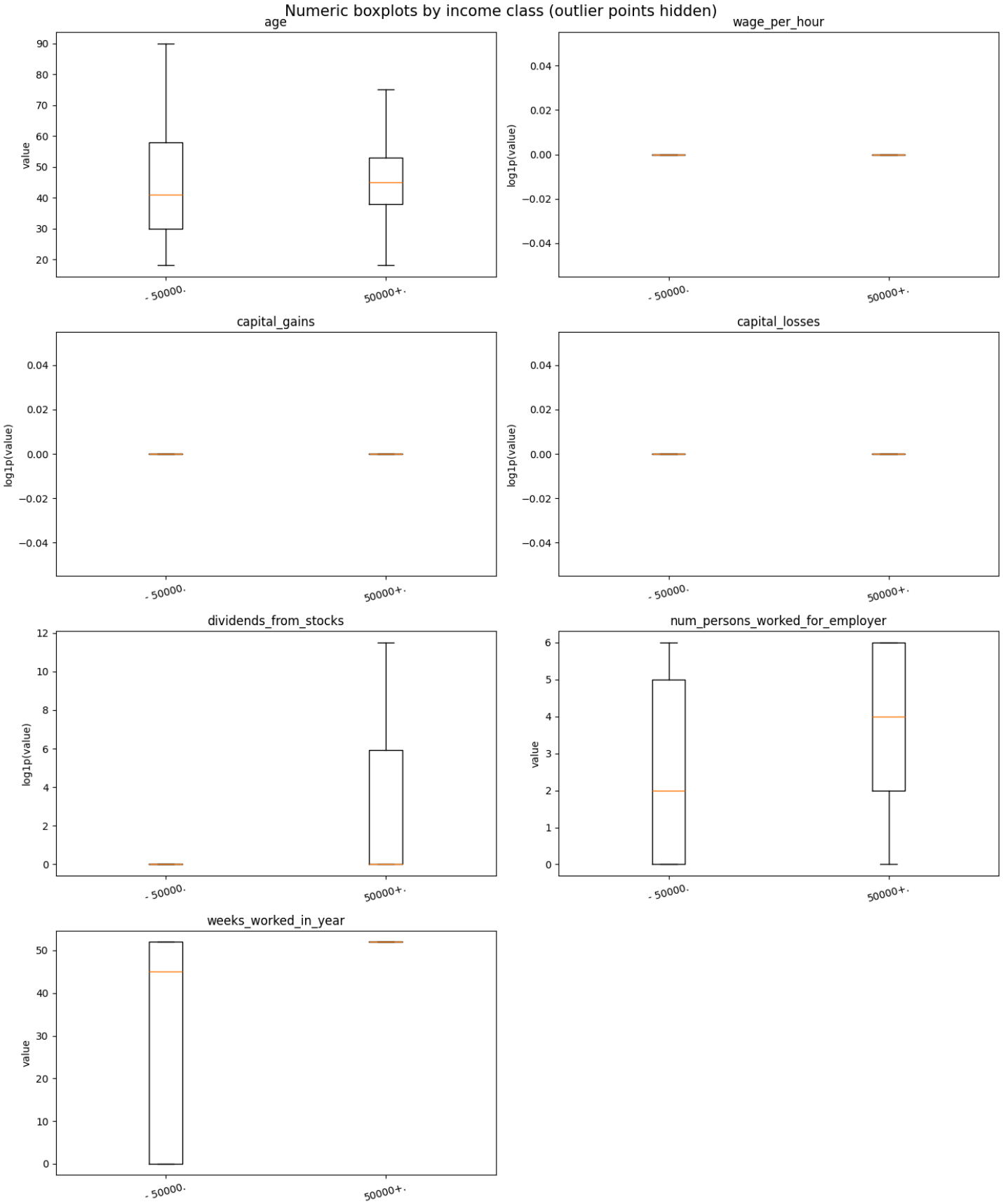


Figure 2. Box plots of numerical features by income class. The four money-related features are shown using a log1p scale, and outlier points are hidden.

Using Pearson's correlation, we found a positive correlation of 0.66 between 'weeks_worked_in_year' and 'num_persons_worked_for_employer'.

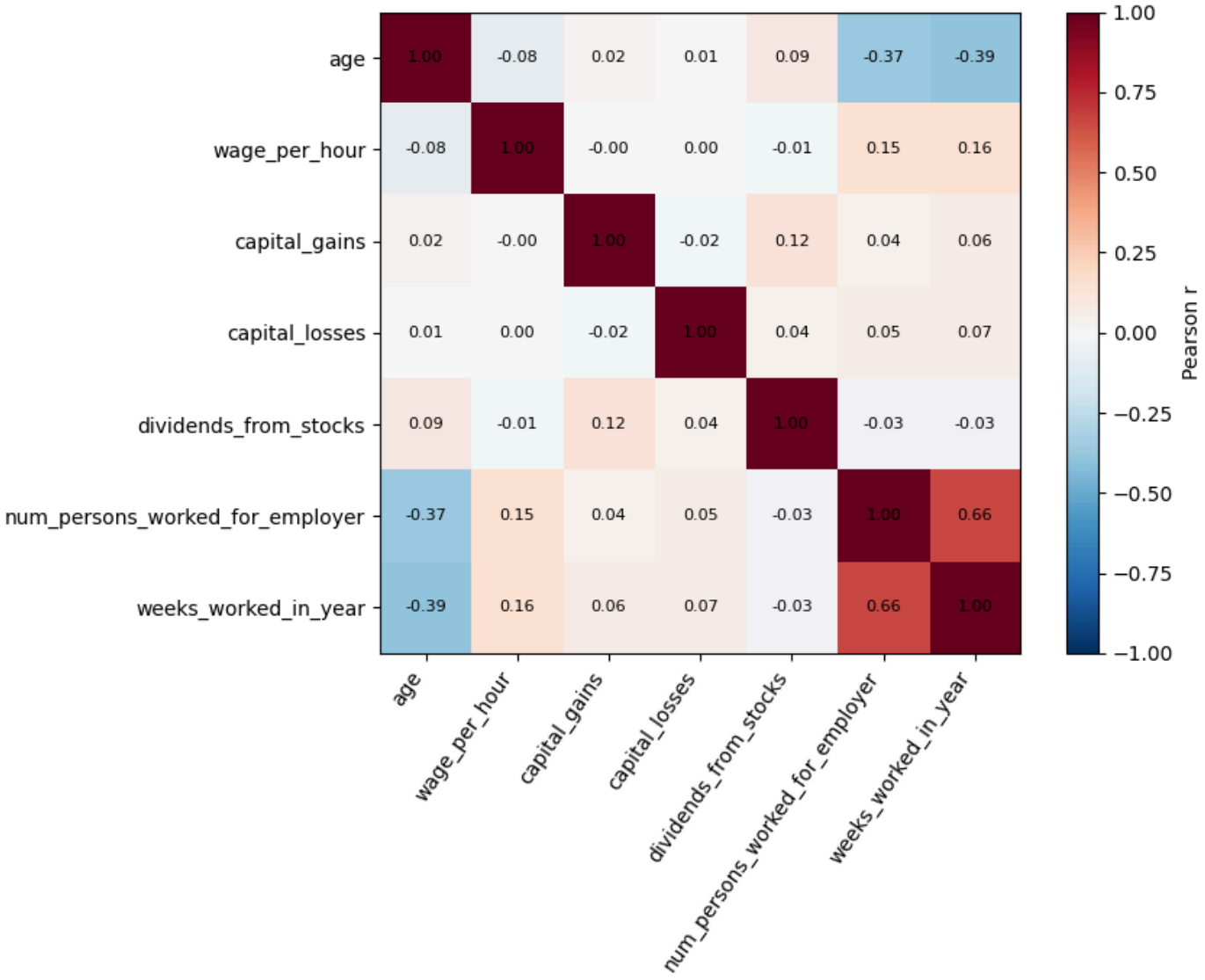


Figure 3. Pearson correlations between numerical features.

For categorical features, we used Cramér's V to examine associations. The pair 'detailed_occupation_recode' and 'major_occupation_code', and the pair 'detailed_industry_recode' and 'major_industry_code' each had a Cramér's V of 1.00, showing a strong association. However, we did not remove features based only on these strong associations.

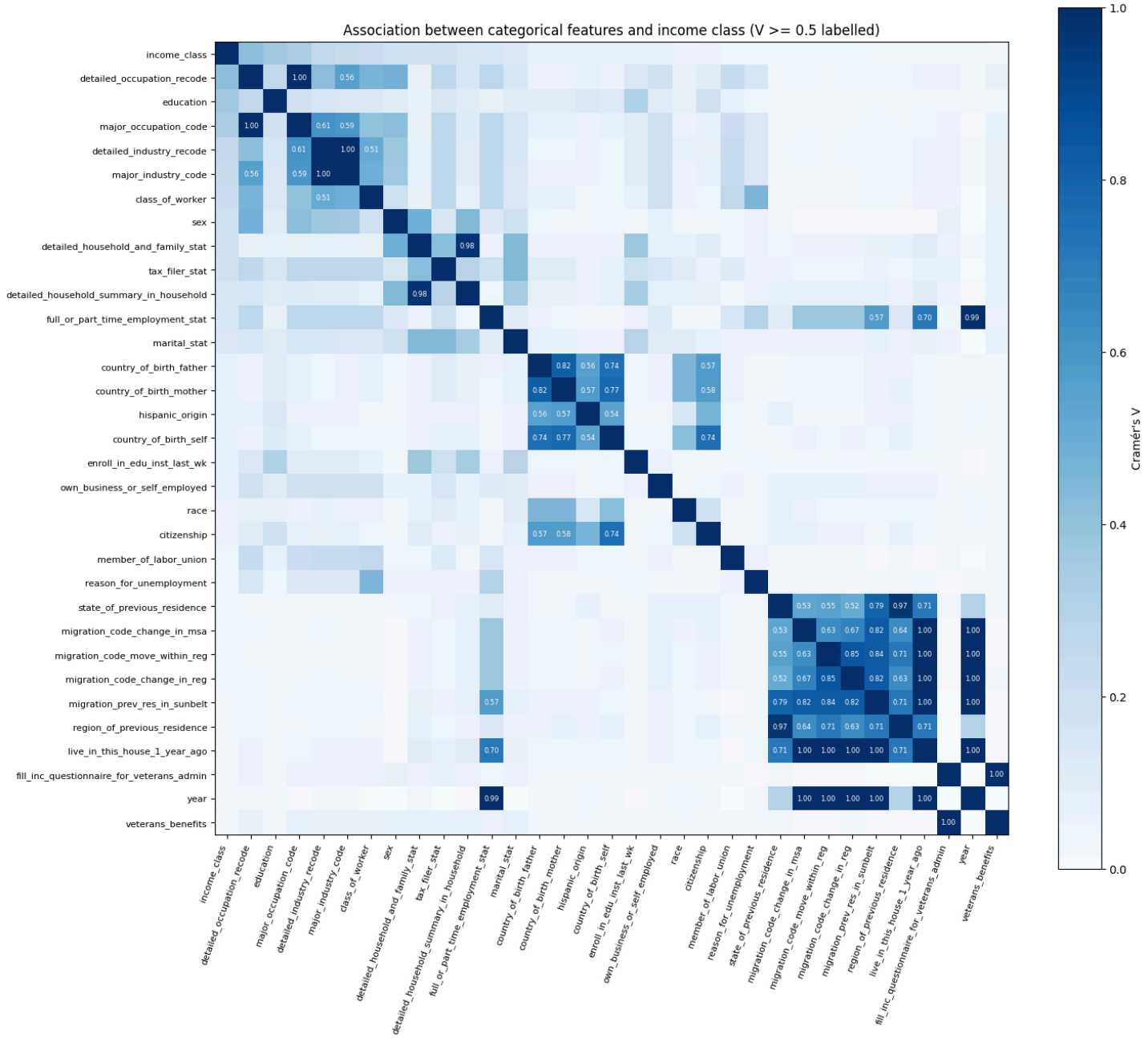


Figure 4. Associations between categorical features and the income class, measured using Cramér's V.

Preprocessing

Data Cleaning

Missing Values

We treated “?” as a missing value and kept “Not in universe” as a category indicating that the question did not apply to the individual. In particular, four features related to migration had 107,895 missing values (50.05%) each. These features were missing in all records from 1995. Since these missing values were related to the survey year, we removed them from the input features. We also removed 'family_members_under_18' because it had the same value for every individual aged 18 or older. Furthermore, we removed ‘live_in_this_house_1_year_ago’ and ‘full_or_part_time_employment_stat’ because each had nearly the same value for most records in one of the survey years. After these steps, the number of input features decreased from 40 to 33.

Outliers

By examining the values of numerical features, we found particularly large values in features related to money, such as ‘capital_gains’, ‘capital_losses’, and ‘dividends_from_stocks’. However, these values are not necessarily input errors and may contain useful information for predicting income classes. For example, the high-income rate among individuals with non-zero ‘capital_gains’ was 32.6%, compared to 7.3% among those with zero ‘capital_gains’, a difference of 25.3 percentage points. Therefore, we did not remove any records based on outliers during this preprocessing.

Data Transformation

For numerical features, we imputed missing values with the median, and then standardized the values using StandardScaler. For categorical features, we imputed missing values as a “Missing” category and applied one-hot encoding. To be able to process categories that did not appear in the training data, we decided to ignore unknown categories. We divided the dataset into 80% for training and 20% for testing using a stratified split that keeps the income class ratio. The training data has 172,459 records, and the test data has 43,115 records. The parameters for imputation and standardization were learned only using the training data. After transforming 33 original features, the full training data now contains 441 columns. To address class imbalance, we also compared models with and without class weights in the preliminary experiment. In this experiment, we use 30,000 samples from the training data and five-fold cross-validation. The AP values below are the averages across the five folds. However, in Logistic Regression, the mean AP without class weights was 0.6048, and with balanced class weights it was 0.5975, so we chose to use no class weights. Also, because money features using original values had a higher mean AP than those with a log1p conversion, we used original values.

Feature Selection

For feature selection, we used Mutual Information (MI), which does not depend on the classification algorithm. MI was computed for the 33 original features using 30,000 samples from the training data, not for the columns after one-hot encoding. This is because we wanted to align with the project objective of comparing the effects of reducing the number of original features. We did not use the test data to compute MI.

The top 5 features with the highest MI scores are listed below.

Feature	MI Score
detailed_occupation_recode	0.0679
major_occupation_code	0.0524
education	0.0474
detailed_industry_recode	0.0334
capital_gains	0.0323

From these results, we found that features such as occupation and education provide relatively more information about income classes. However, because MI evaluates each feature individually, this method does not account for duplicate information among the features. Following the requirements, we chose the top 17 features as a candidate feature set based on the MI ranking. The feature set is about half of the 33 original features. The reason we divided the feature sets is that the value of MI drops almost by half between the 17th and 18th features. We did not use any classification methods during this selection. Also, we still keep the 33 original features. We will finalize the feature set after comparing the prediction performance and computation time of both feature sets in the classification experiment. In the experiment, we will re-calculate MI using only the training part of each fold of cross-validation.

Dimensionality Reduction

To check the high-dimensional data after one-hot encoding, we applied PCA. Using 30,000 samples extracted from the training data, we found that 63 principal components were needed for the 440 columns after transformation to reach a cumulative explained variance ratio of 90%. Also, we made scatter plots by income class and survey year using the first 2 principal components.

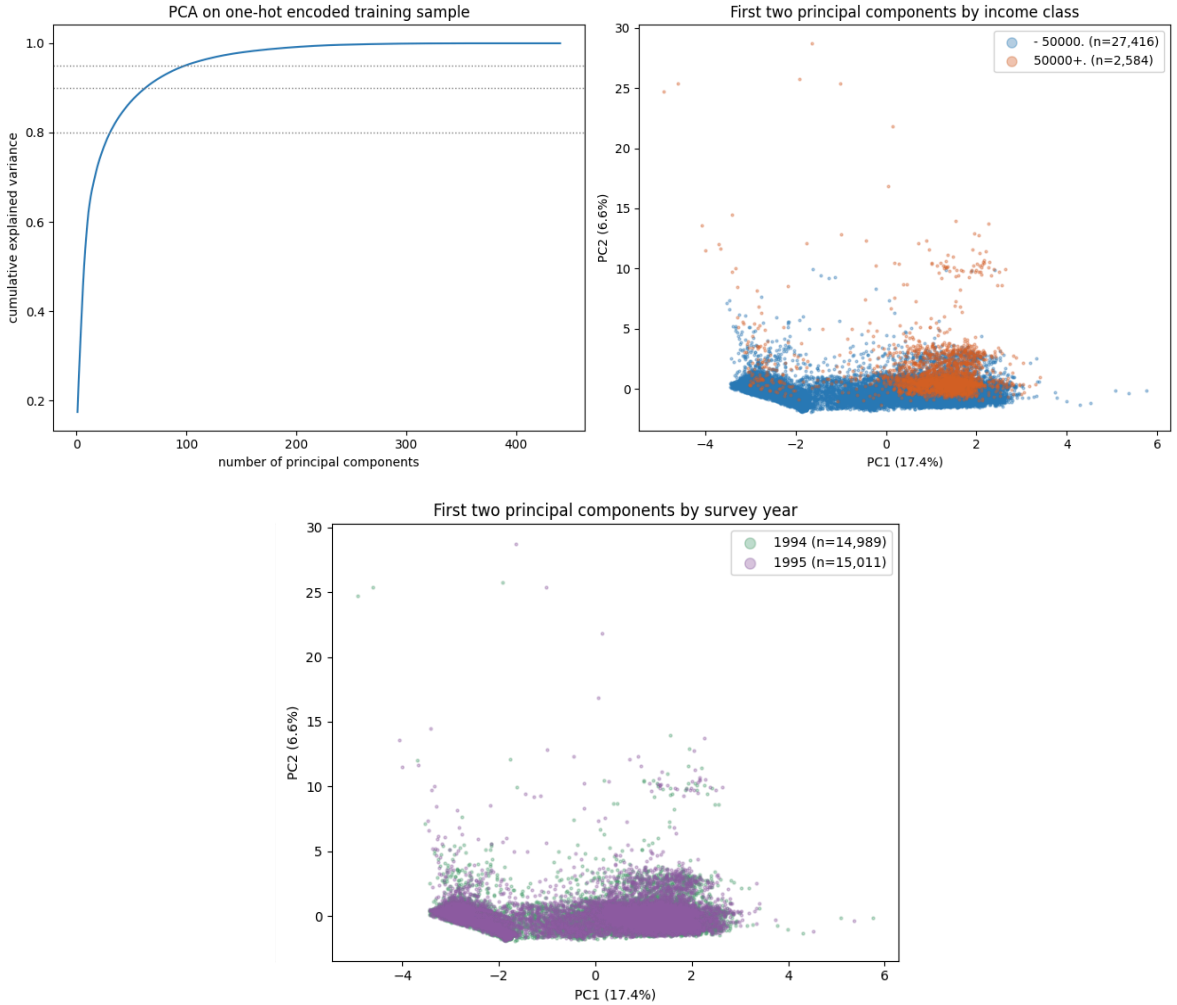


Figure 5. PCA results for 30,000 training samples: cumulative explained variance and scatter plots of the first two principal components by income class and survey year.

PCA in this case was used to check the data structure, and we decided that we will not introduce PCA as default preprocessing for the classification model.